

浙江大学



Lab Practice Report for Undergraduate

Name:

College:

ZheJiang University

Department:

Science

Major:

Marine Science

Student NO.:

Instructor:

Yanan Di

Date: 2024.12.6

Lab Practice Report of Zhejiang University

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Course Name: Marine biology Type of Practice: Analyse

Name of Practice: Quantitative confirmation of microalgae

Student Name: _____ Major: Marine biology

_____ Student Number:

Team Member: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.10.10

1. Aims and Requirement of the Practice (Compulsory)

- (1) Learn the basic procedure to measure catalytic activities in biological samples
- (2) Understand the basic theory of Ellman's assay
- (3) Measure the activity of AChE in the biological samples from different types of circulations system
- (4) Compare the activity in blood/haemolymph from different organism and analyse the results with their circulation characteristics.

2. Background information of the Practice

Environmental pollutants such as organophosphorus can cause serious harm to wildlife and humans. The measurement and interpretation of acetylcholinesterase (AChE) activity in aquatic organisms is therefore topical.

AChE is a serine hydrolase mainly bound at neuromuscular junctions and cholinergic brain synapses. AChE hydrolyzes choline esters. The principal biological role of AChE is termination of impulse transmission at cholinergic synapses by rapid hydrolysis of the neurotransmitter,

acetylcholine (ACh). The powerful acute toxicity of organophosphorus poisons is due primarily to the fact that they are potent irreversible inhibitors of AChE, forming a covalent bond with a serine residue at the active site.

The AChE assay is based on the Ellman assay in which thiocholine, produced by AChE, reacts with 5,5'-dithiobis (2-nitrobenzoic acid) (DTNB) to form 5-thio-2-nitrobenzoic acid (TNB), a colorimetric (412 nm) product, proportional to the AChE activity present

3. Equipment and materials

- (1). Supernatant from blood samples you have preserved before
- (2). Freezing Microcentrifuge, Microplate reader and 96-well microtiter plate
- (3). Water bath (37 °C), floating racks, timer, Pipette (1 ml, 200 µL, 10 µL) and tips with different sizes
- (4). Microcentrifuge tubes (1.5 ml) and racks, Sterilized test tube (50 ml), glass beakers (100 ml) for discarding purpose
- (5). Rabbit blood sample (2%) (R), PBS, distilled water (B1), Solution 1 (S1), Solution 2 (S2), Solution 3 (S3), Solution 4 (S4)
- (6). Marker pens, pencils, paper towels, plastic bags, gloves, ice box

4. Procedures

(1) Sample preparation

① Rabbit blood sample: Mix rabbit blood with lysing solution (1:1). Divide it into 3 microcentrifuge tubes (200 µL), mark the tube and gently homogenize and centrifuge at 4 °C 10000g 5 min.

② Take out preserved fish blood extraction and melt on ice.

③ Use PBS as blank. 3 replicates

(2) AChE measurement

Sample:

① add 50 µL S2 into 15 µL sample, react in microcentrifuge tube and put in water bath for 5 min.

② Add 50 µL S4 to sample, mix thoroughly centrifuge at 12000 rpm, 5 min at room temperature. Gently take out 10 µL of supernatant from each tube and transfer into a well

in 96 microplate

③Add 170 µL S1 and 20 µL S3 into sample

④ Gently mix with repeated pipetting, please avoid the bubbles. Leave at room temperature for 2 min and measure all the samples by microplate reader. Make a record. Calculate the change of absorbance

Blank:

①15 µL blank react in microcentrifuge tube and put in water bath for 5 min.

②Add 50 µL S4 and 50 µL S2 to sample, mix thoroughly centrifuge at 12000 rpm, 5 min at room temperature. Gently take out 10 µL of supernatant from each tube and transfer into a well in 96 microplate

③Add 170 µL S1 and 20 µL S3 into sample

④ Gently mix with repeated pipetting, please avoid the bubbles. Leave at room temperature for 2 min and measure all the samples by microplate reader. Make a record. Calculate the change of absorbance

(3) Calculation of AChE activities

$$\text{AChE (U/mL)} = (\Delta A \div \epsilon \div d \times V_{\text{coloured}} \times 10^9) \div (V_{\text{sample}} \times V_{\text{supernatant}} \div V_{\text{catalyse}}) \div T_{\text{reaction}}$$

$$K = 3758.170.$$

5. Questions

(1) What S1-S4 means

S1: Buffers

S2: Ach

S3: DTNB

S4: Buffers

(2) Calculate the AChE activity in each sample; please pay attention about ALL dilutions

The resulting is shown in the table below

	1	2	3	average	ΔA	AChE (U/mL)
鱼上清	0.1508	0.1502	0.1447	0.1486	0.0020	88.6928
鱼全血	0.1672	0.1689	0.1736	0.1699	0.0233	1050.7843
兔血	0.1604	0.1645	0.1643	0.1631	0.0165	18565.3598
空白	0.1532	0.1492	0.1374	0.1466		

(3) Discuss the results and analyse the reasons of such results, pay attention to different circulation systems.

The AChE content of rabbit blood was significantly higher than that of fish blood

Reason:

Closed Circulatory System: The rabbit's closed circulatory system can more efficiently transport oxygen and nutrients, while requiring a more robust neural regulatory mechanism to maintain stable blood pressure and blood flow.

Efficient Heart Function: The four-chambered heart can efficiently separate oxygenated and deoxygenated blood, ensuring efficient circulation of blood between the lungs and other parts of the body.

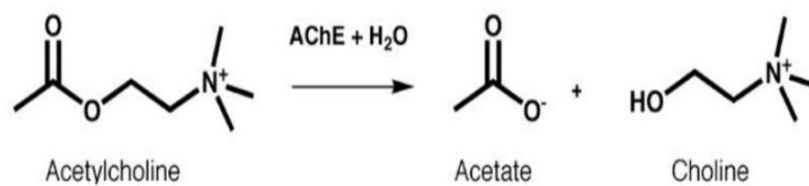
High Blood Flow and Pressure: In a closed circulatory system, blood flows quickly and at high pressure, necessitating a more powerful neural regulatory mechanism to maintain stable blood pressure and blood flow.

(4) Two blanks have been provided in the practice. Explain the reasons about your selection.

In this experiment, both PBS and LYS were used as buffers. My choice is PBS.

Because PBS does not contain any enzymes or other active substances on its own, so it does not interfere with the activity of AChE and can provide a very clean background signal. Besides, PBS is simple in composition and free of various chemicals in the lysate, non-specific interferences can be reduced.

(5) There are two basic biochemical reactions involved in the AChE assay. Please list the potential equations of the reactions.



(6) Attach your calculation procedure about K in the report.

$$\begin{aligned}
 K &= \frac{V_c \times 10^3 \times V_{ys}}{\varepsilon \cdot d \cdot V_{sa} \cdot V_{\text{电泳}} \cdot T_{\text{run}}} \\
 &= \frac{0.2 \times 10^3 \times 0.115}{13.6 \text{ mm}^2 \text{ cm}^2 \times 0.6 \times 0.015 \times 0.01 \times 5} \\
 &= 3758.170
 \end{aligned}$$

注：不同类型的实验课对实验报告可有不同要求，各个学科的实验报告可以根据自己的学科特点做适当的调整，但上述基本内容中的第一、二、三、六条为必须填写的内容。